

```

1  %-----
2  % Consumer Choice
3  %-----
4
5  clear
6  clc
7
8  %Printing the outputs as a text file if ==1
9  output_file=0;
10
11 %Problem Set-up
12 alpha=0.7;
13 fun=@(x) -(x(1).^alpha).*(x(2).^(1-alpha)); %minimizing (-u)
14
15 A=[1, 1]; %A=[p1, p2]
16 b=[1]; %b=[m]
17 Aeq=[];
18 beq=[];
19 lb=[0; 0];
20 ub=[];
21 nonlcon = [];
22 x0=[0.1; 0.1];
23
24 %fmincon
25 [x,fval,exitflag,output,lambda,grad,hessian] = fmincon(fun,x0,A,b,Aeq,beq,lb,ub,nonlcon);
26
27 if output_file==1
28     dfile="consumer_choice_fmincon.txt";
29     if exist(dfile, 'file')
30         delete(dfile);
31     end
32     diary(dfile)
33     diary on
34 end
35
36 %Output
37 fprintf('alpha=%.3f, p1=%.3f, p2=%.3f, m=%.3f\n',alpha,A(1),A(2),b);
38 fprintf('Optimal consumption: %.3f,%.3f\n',x(1),x(2));
39 fprintf('Utility: %.5f\n',-fval);
40 fprintf('Multiplier (budget const): %.5f\n',lambda.ineqlin);
41 fprintf('Multiplier (non-negativity const): %.5f, %.5f\n',lambda.lower(1),lambda.lower(2));
42
43 if output_file==1
44     diary off
45 end
46
47 %Plot
48 x1_vec=linspace(0, b/A(1), 100);
49 x2_vec=linspace(0, b/A(2), 100);
50 [x1_mesh, x2_mesh] = meshgrid(x1_vec, x2_vec);
51 figure();
52 hold on;
53 grid on;
54 plot(x1_vec, (b-A(1)*x1_vec)/A(2),'k', 'LineWidth', 1); %budget constraint
55 contour(x1_mesh, x2_mesh, (x1_mesh.^alpha).*(x2_mesh.^(1-alpha)), [(-fval), (- fval)],
56 'LineWidth', 1, 'LineColor', 'b');
57 contour(x1_mesh, x2_mesh, (x1_mesh.^alpha).*(x2_mesh.^(1-alpha)), [0.75*(-fval), 1.25*(-fval)],
58 'LineWidth', 1, 'LineColor', 'k', 'LineStyle', '--');

```

```
57 plot(x(1),x(2),'ro','LineWidth', 1);
58 xlabel('$x_{1}$','Interpreter','latex','FontSize',12);
59 ylabel('$x_{2}$','Interpreter','latex','FontSize',12);
60 title('Consumer Choice','Interpreter','latex','FontSize',14);
61 xlim([0,b/A(1)]);
62 ylim([0,b/A(2)]);
63
```